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An Innovative Smart Multistage Technology Adopted to Address Challenged Completion and Acid Fracturing Stimulation Methods in High Temperature Formation Wells Through Advanced Dissolvable Technology

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Abstract

Over the last few years, efforts have been made to improve the multi-entry multistage fracturing (ME MSF) system to increase reservoir contact within a single wellbore, enhancing hydrocarbon recovery. However, longer lateral lengths are required to raise stage counts, normally constrained by available technologies. While innovations have been introduced, they often yield mixed results in terms of reliability and predictability. A new approach is needed to overcome these limitations into existing operations without introducing additional risks.

An inventive ME MSF technology was developed to address these limitations, enabling longer laterals, increasing stage counts while improving stimulation efficiencies and derisking operations. This technology eliminates the risks and costs associated with wellbore milling post-stimulation. It incorporates unique intelligent, dissolvable darts that self-identify specific sleeves before activating to provide zonal isolation. These darts are flexible enough to open either single or multiple sleeves with a single deployment. The primary goals were to increase reservoir contact, enhance operational efficiencies, and reduce risks, making this technology a significant advancement in the field.

In a high pressure, high temperature (HPHT) candidate well situated in a carbonate reservoir, extreme conditions were encountered, with high bottom hole temperature (BHT) and requiring a 15,000-psi applied pressure differential for acid fracturing stimulation. To address these challenging conditions, the ME MSF technology underwent significant adaptations from its original design parameters while retaining its unique features. The system's outer diameter was constrained to a maximum of 5.25" for the 4.5" ME MSF, ensuring it met or exceeded the liner string's burst and collapse ratings. Additionally, the technology maintained the capability to rotate the liner string into the wellbore and function within its original parameters. A specially designed intelligent dissolvable dart was employed to manage the challenging conditions, capable of maintaining a 15,000-psi differential pressure at high reservoir temperature for up to 5 hours under extreme low pH environment, while ensuring zonal isolation for at least 12 hours at a reduced pressure

differential. This adaptation exemplifies the technology's versatility and robustness in handling extreme operational demands.

This paper will review the design optimization, rigorous testing conducted to replicate downhole conditions as closely as possible, and the deployment of the system. It will also cover the execution of the acid fracturing stimulation, improved operational efficiencies, a normalized hydrocarbon production comparison with offset wells, and the outlook for the unique ME MSF system

Introduction

As gas reservoirs continue to play a vital role in the global energy landscape, the limitations of conventional completion technologies are becoming increasingly evident. Elevated reservoir temperatures and higher fracturing pressures pose significant operational challenges exposing the limitations of legacy completion technologies to fully unlocking the potential of these wells. Traditional completion methods, such as plug-and-perforate (PnP) systems, are showing limitations in these extreme downhole conditions, which is exasperated in wells with extended lateral lengths, while maintaining cost-effectiveness.

The ongoing drive to reduce costs and maximize in both conventional and unconventional plays, are pushing completion technology providers to innovate. In response, new disruptive technologies have emerged, such as unlimited fracturing sleeve systems, that aim to improve horizontal multi-stage completions. While PnP and ball drop systems remain widely used, each presents operational tradeoffs. One persistent challenge across all systems is the need for operational flexibility when unexpected issues arise during drilling, completion or stimulation phases of the operations.

To address these challenges, a new fracturing sleeve system was developed and deployed. Engineered to reduce operational complexity and risk, this system delivers uniform fracture stimulation throughout the lateral. It allows for an unlimited number of fractured stages, supports continuous stimulation, and eliminates the need for any well intervention post stimulation. Using identical sleeves for every stage simplifies liner installation, streamlines logistics and improves inventory management. Additionally, the sleeves simplified internal geometries with tungsten carbide nozzles to mitigate erosion ensuring even flow distribution of the stimulation fluid across all the nozzles within a stage. This uniform fluid placement promotes more consistent fracture propagation in heterogeneous reservoirs for enhanced reservoir production.

Innovative Multi-Entry Multistage Fracturing System

The most cost effective and efficient method to complete any tight gas well is to increase the contact area of the formation by drilling horizontally and having multiple orthogonal hydraulic fractures along the liner. At present the preferred method to obtain this configuration is the plug & perforation method which provides multiple entry points per stage. This method is where a wireline unit conveys a frac plug to provide the necessary isolation from the open zones below and perforating guns to provide the necessary access to the formation from the liner. The plug & perf method is a mature, off the shelf technology but some underlying factors have identified the need for another completion method. With the repeated wireline interventions, the downtime between stimulation intervals, the failure rate of the frac plugs and the required deployment of coiled tubing to perform either a drill out of the frac plugs or to perform a wellbore cleanout to allow the well to flow is becoming a hinderance to the overall cost to complete a well as the laterals are getting longer. Some other factors influencing the need for another completion method include optimizing pumping efficiencies for single wells, simplification of cluster spacing/grouping, regions with casing deformation concerns, the desire for extended reach wells and or the elimination of staggered ball drop systems (E.R. Davis et al., 2023).

As it is observed by Arias et al. (2022) and Rodoplu et al. (2022), the MSF ball drop technology has been an effective alternative to PnP technique and has evolved significantly over the past decade, providing solutions to meet well completion requirements. This evolution has addressed increasing pressure

differential requirements, introduced advancements in dissolvable technology, and expanded the application of cemented MSF systems, imitating the typical cluster design from PnP. However, when the application environment complexity increases, for example, acid fracturing operations at high temperature formations, the aforementioned system presents limitations, particularly in high-pressure environments with a 15,000 psi differential. Firstly, the operational lower completion ID requirement range of 2.5 to 3.5 inches imposes a restriction on the number of frac stages that can be effectively implemented, limiting the system's scalability and adaptability in complex or extensive reservoirs. Secondly, the dissolvable ball technology faces significant challenges in harsh operational conditions, such as high temperatures, relative low mass and low-pH environments encountered during acid fracturing operations. These conditions can compromise the integrity and reliability of the balls, affecting the overall efficiency and consistency of the fracturing process. Additionally, the time sensitivity of the dissolvable balls adds another layer of complexity, as precise control over their degradation rate is crucial to avoid operational delays or incomplete fracturing, which could impact the well's productivity. Lastly, the staggered ball baffles in the system create a variable ID profile, necessitating the use of custom-made wiper plugs to ensure proper clean-out of cement in case of cemented MSF technology. This requirement increases the system's complexity and costs, introducing logistical challenges in sourcing and deploying these specialized components. These limitations collectively highlight the challenges of employing MSF ball drop technology in high-pressure, challenging environments, underscoring the need for careful consideration and alternative solutions in such scenarios.

A step change in the multi entry multistage fracturing (ME MSF) system was developed to address some of the mentioned underlying factors as well as to convey a reduction in completion complexities, derisk operations, provide the ability to increase reservoir contact area, thereby increasing well productivity. The innovative ME MSF system was advanced to eliminate wireline deploying frac plugs with perforation guns and to eliminate coiled tubing operations.

The innovative ME MSF system consists of Intelligent dissolvable darts (IDD) that are programmed on surface and pumped to the target frac sleeve to be stimulated. These darts are a significant improvement over ball drop or mechanical dart systems as they recognize the innovative (ME MSF) sleeves without ID baffle restriction and without physical contact for index. This enables them to be pumped through the well at higher rates. The IDD opens the required sleeve/s before landing into a sleeve that provides the required isolation for the stage frac. Once the frac has been pumped these darts dissolve to provide near full bore ID thus allowing the well to flow without the need for any coiled tubing intervention. The installation of these frac sleeves is done when the casing/liner is deployed in the well. Each frac sleeve has an inner sleeve that is held in place with shear screws which is opened when an activated dart engages the sleeve. When the activated dart is engaged in the sleeve, pressure is allowed to increase creating a differential pressure which provides the necessary force required to shear the pins and shift the inner sleeve open. Once the inner sleeve is in the open position a flow path is exposed from inside the liner to the reservoir and the zonal stimulation can proceed.

The innovative MSF system can be configured for single point, one dart opening one sleeve to provide the required isolation for that stage or a limited entry (LE) cluster system [Figure 1](#), one dart opening multiple sleeves before landing in an isolation sleeve [Figure 2](#) or a combination of both systems in a single wellbore. Further variation for the innovative MSF system is either configuration can be deployed in an open hole or in a cemented wellbore.



Figure 1—Limited Entry Cluster Sleeve



Figure 2—Limited Entry Landing Sleeve

With a LE cluster system there are two types of frac sleeves, the cluster sleeve and a landing sleeve which provides the required isolation for that stage. In a LE MSF cluster system, one or more cluster sleeves are installed above a landing sleeve to define the stage and is opened using an activated IDD during the pumping operation. When an activated IDD engages a cluster sleeve it temporarily provides the required isolation to enable pressure to be built up and shear open the inner sleeve. The opened cluster sleeves do not retain the activated dart only impedes its movement, thus allowing it to travel down hole to the next LE sleeve. Fluid is prevented from leaking out the limited entry tungsten carbide nozzles in any opened LE cluster sleeve by shearable discs, thus retaining the required fluid to keep the IDD moving downhole. If the IDD encounters another LE cluster sleeve, the process is repeated until it encounters a LE landing sleeve which prevents the IDD from travelling further. The landing sleeve can include tungsten carbide nozzles and has an internal profile that enables the dart to isolate the previously opened clusters sleeves in that stage.

Each IDD launched into a well is referred to as unactivated and is programmed on surface to identify the unique MSF sleeves and count these sleeves as it is travelling through the well to reach its target sleeve. Once the pre-programmed sleeve count is recognized, the dart activates to increase its operational outside diameter (OD), thereby allowing it to engage into the target sleeve in a single point system or sleeves with a cluster system. This process of dropping pre-programmed unactivated IDD's into the wellbore to their programmed target sleeves, is repeated until all the sleeves in the wellbore have been shifted open and stimulated. The operational OD of the unactivated IDD's in conjunction with its short length is ideal for wells in areas prone to casing deformation.

The innovative LE MSF system is a dramatic improvement to the plug & perf completion system. Each cluster sleeve has manufactured tungsten carbide nozzles with machined ID's instead of perforation holes with a diameter's dependant on proper centralization of the perforating gun carrier. Erosion of the perforation diameter in the casing/liner, which has a Rockwell hardness (HRC) ranging from 26-36 depending on the material, is quite common and leads to a reduction in cluster efficiency (Long et al. 2017). Having a uniform, machined ID in-conjunction with hardened tungsten carbide nozzles (HRC ranging from 70-80) provides a consistent flow path throughout the zone with a predictable pressure drop across all the nozzles to ensure even flow distribution for the stimulation fluid. This dramatically reduces starved verses dominant zones leading to better overall fractured stages for better production. Another detriment to the plug & perf system is the frequency of a frac plug unexpectedly slipping or failing and the damage these failures are doing to the casing/liner (Jacobs JPT article 2021). Any frac plug failures or uneven stimulation fluid distribution is further exasperated in extended lateral wells. The innovative ME MSF system alleviates these failures as the IDD isolates into manufactured sleeves and any failure can be resolved by programming another IDD to land into that sleeve to continue with the frac without damaging the casing/liner.

An additional key factor of the innovative ME MSF system is that it is not limited by the operational depth of wireline, nor by the limitations of coiled tubing, or the number of stages as with a graduated ball seat

system. Each isolation sleeve in the innovative ME MSF system has the same internal diameter (ID), and the IDD can be configured for an unlimited number of stages in a well. Since any IDD can be programmed to engage with the sleeve, extreme lateral lengths can be drilled regardless of the well depth or lateral length.

Adapting Technology to High Pressure & High Temperature

Well completion companies have historically engineered their multi-stage fracturing (MSF) technologies to operate under pressure differentials of up to 10,000 psi and temperatures ranging from 180 to 210°F. These specifications were well-suited to the shallower and cooler reservoirs commonly encountered in earlier unconventional developments. However, the growing prevalence of horizontal wells with extended lateral sections has led to an increase in the number of completion components required per well. This escalation raises the risk that the failure of even a single component can compromise operational efficiency and cost-effectiveness. To address this challenge, it is essential for technology providers to establish a robust track record of successful tool deployments to validate new technologies before introducing them into more extreme downhole environments characterized by greater depths, elevated temperatures, and higher pressures. Any system with modified specifications must undergo extensive testing under simulated field conditions to ensure performance reliability prior to field implementation.

As trend's push into tighter formations that demand greater stimulation pressures and higher temperature tolerances, continued modification of legacy completion systems such as plug-and-perf has become increasingly impractical. To ensure reliable performance in these more demanding tight gas reservoirs, it has become necessary to raise the operational pressure capability to 15,000 psi and the temperature rating to 330°F.

In response to these challenges, the ME MSF system was designed with specific enhancements. These included reducing the sleeves outer diameter from 5.5 inches to 5.25 inches, increasing the burst and collapse pressure ratings to 15,000 psi, and maintaining the ability to rotate a 4.5-inch liner into the wellbore during drilling if needed. The IDD was similarly engineered to meet the demanding specifications, with a differential pressure rating of up to 15,000 psi, temperature tolerance up to 330°F, and sustained operational integrity for at least six hours in high-acid concentrations during the stimulation without degradation in performance. It was also designed to dissolve in a controlled manner, allowing for well flowback without the need for coiled tubing intervention. Furthermore, the ME MSF system retained operational flexibility for both cemented and open hole applications, eliminating the need for alternate sleeves or on-site modifications during drilling.

Sleeve Validation & Testing

The initial step involved evaluating the current ME MSF sleeve design followed by a detailed assessment of any potential modifications to individual components and other related elements of the ME MSF system. Particular focus was placed on internal sleeve components, as any changes to these parts could have cascading effects on the system. This review also examined how such modifications might impact component integration and overall performance ensuring all changes align with the defined operational requirements.

The primary engineering requirement for the ME MSF sleeves was to achieve a burst/collapse rating of 15,000 psi, along with a reduction in outer diameter from 5.5" to 5.25", while maintaining the original profile ID. A secondary requirement was to preserve the sleeve's ability to be rotated into the wellbore if needed during the drilling operations. These combined requirements initiated a comprehensive design review by the engineering team.

Following the design review, revised sleeve components were manufactured and assembled along with the corresponding elastomers for HPHT application into their respective sleeves to be validated tested. The

first test was to apply 15,000 psi internal pressure at 330°F to monitor for any leaks with the revised sleeve components. After 45 minutes no leaks were visible, the test was deemed to be successful as per Figure 3.

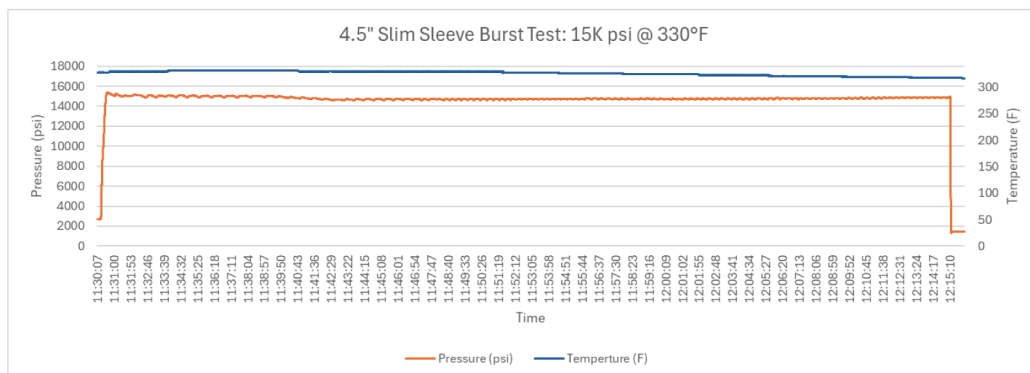


Figure 3—Slim Sleeve Burst Test

The second test involved internal cyclic testing by applying 15,000 psi pressure at 330°F, to further assess the durability and performance of the revised components for the sleeve (Figure 4). The next two tests were conducted to apply exterior pressure to validate the collapse resistance of the sleeves. The final test involved external pressure on the sleeve to simulate field conditions and further verify performance under operating conditions. Figure 5

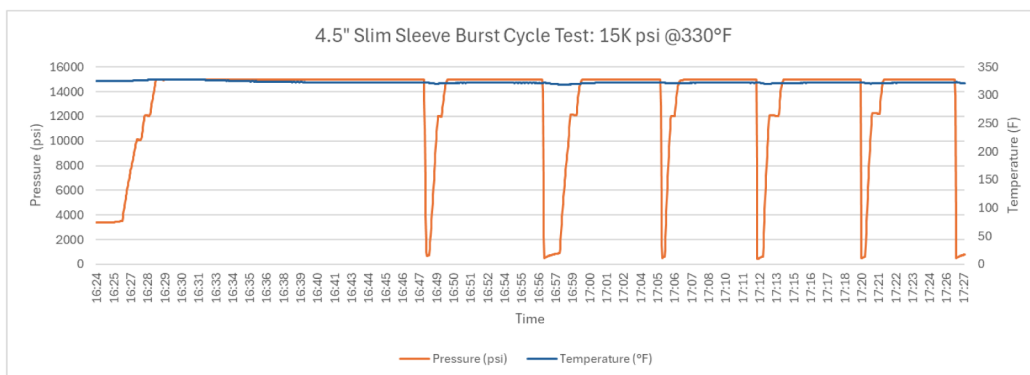


Figure 4—Slim Sleeve Cycle Test

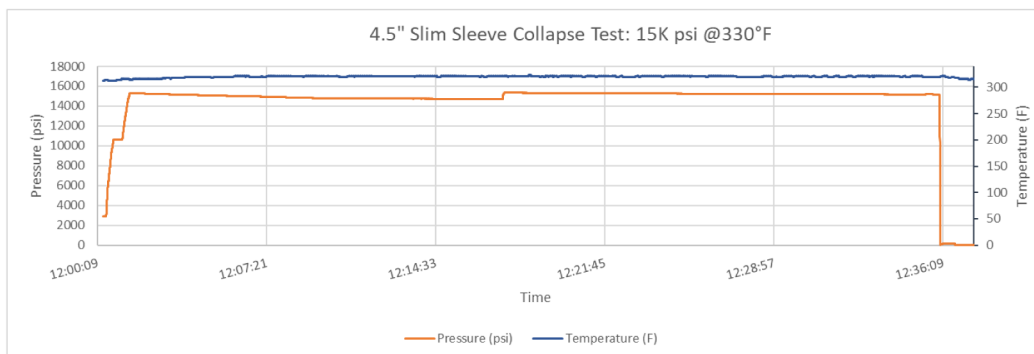


Figure 5—Slim Sleeve Collapse Test

Intelligent Dissolvable Darts

The core innovation of the ME MSF system is the intelligent dissolvable darts. These small compact unactivated darts are programmed on surface to count the innovative ME MSF sleeves. Immediately after

the programmed count has been recognized, the IDD activates to increase its operational OD allowing them to engage into the next ME MSF sleeve. After finalizing the ME MSF sleeve design, the engineering team began a detailed review of the intelligent dissolvable darts (IDD) showed in Figure 6.



Figure 6—Intelligent Dissolvable Dart

The updated operational requirements for the IDD's included an increase in differential pressure rating of 15,000 psi and an increase temperature rating of 330°F, while maintaining the dart's ability to dissolve within a timeframe compatible with standard acid fracturing flowback operations. To meet these enhanced specifications, various proprietary materials were investigated and tested to evaluate their suitability under the higher pressure, higher temperature, in a heavy brine fluid for acid stimulation operations.

Two proprietary materials were developed and selected for testing in high-pressure, high-temperature and in high brine concentration environments. The first step involved testing the materials in fresh water at ambient temperature to observe the darts performance. This initial test aimed to assess how each material would respond to the force generated at the elevated pressure. Proprietary Material 1 was chosen first for initial testing.

During the test, pressure was gradually increased in roughly 2-minute intervals, until 15,000 psi was achieved. After applying 15,000 psi, a small 200 psi pressure drop was observed above the dart. As pressure was reapplied, it continued to drop. Due to the pressure continuing to drop the test was terminated, and the test fixture was disassembled for inspection. Figure 7

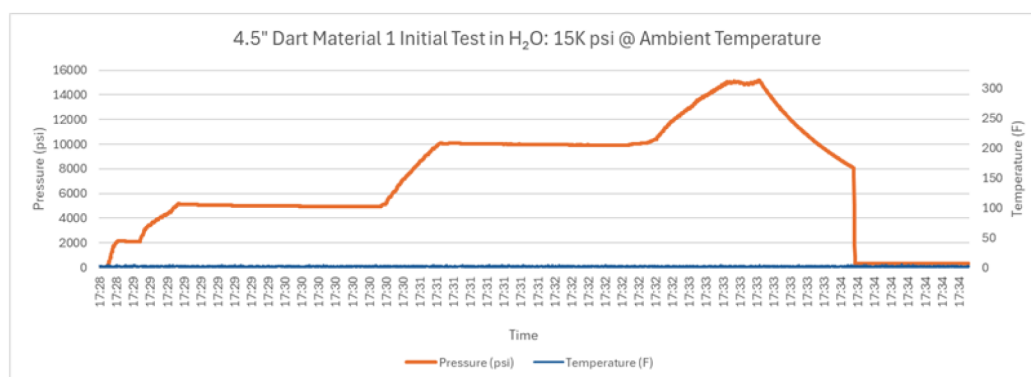


Figure 7—4.5" Dart H₂O Material 1 Test

Upon inspection the pressure drop was due to a leak in the test fixture equipment and not the material. After repairing the test fixture, the test was repeated. Another IDD was assembled with proprietary material 1 and placed in the test fixture, ensuring it was fully submerged in freshwater throughout the test. Pressure was applied in two steps to 14,750 psi after the temperature stabilized. The pressure held 14,500 psi pressure

for approximately 5 hours, whereby it was arbitrarily terminated. The test validated the IDD could withstand elevated pressures at 330°F for extended periods of time in fresh water. [Figure 8](#)

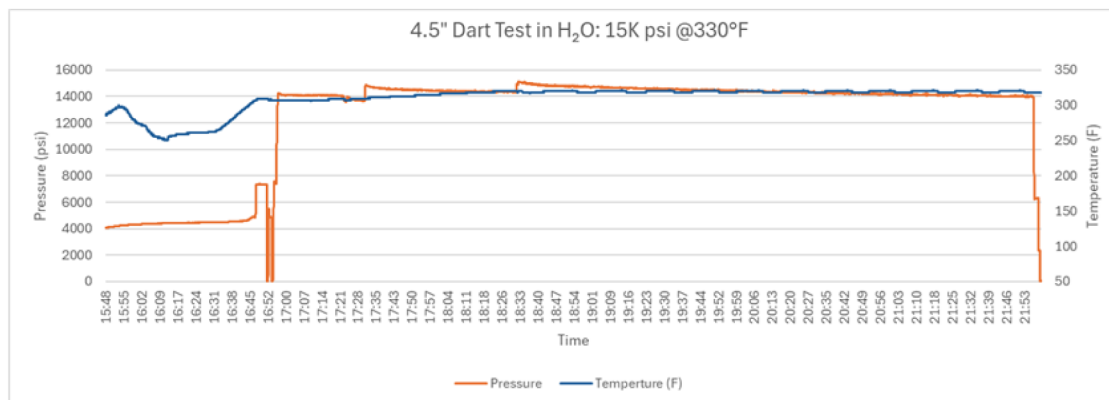


Figure 8—4.5\" Dart H₂O Extended Test Material 1

The test apparatus was disassembled with the IDD being thoroughly inspected. A second test was conducted with 3% KCl fluid. The IDD was placed into the fixture fully submerged in the solution and heated to 330°F. Pressure was applied above the dart to 15,000 psi, once the temperature stabilized. The dart held 15,000 psi for nearly 5 hours before the pressure began to slowly decrease. A few minutes after the initial pressure decrease, the pressure rapidly dropped and the test was terminated. The total amount of time the dart withstood 15,000psi at an elevated temperature of 330°F in 3% KCl was nearly 5 hours. Due to the pressure decrease and the requirement for extended period of the activated IDD to be on seat, the test was deemed an unsuccessful. [Figure 9](#)

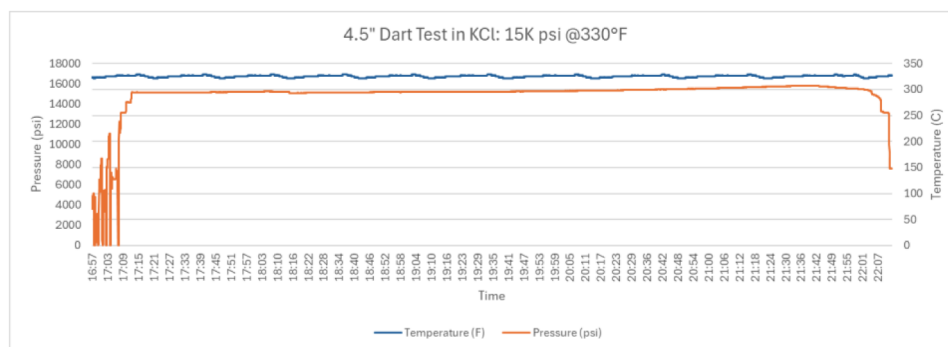


Figure 9—4.5\" Dart KCl Test Material 1

Upon further inspection it was determined material 1 was unable to withstand 15,000 psi at 330°F in 3% KCl fluid for a prolonged period of time. Based on the findings another IDD was assembled with proprietary material 2 and inserted into the test fixture. A heavier brine mixture of 150,000 ppm KCl with a pH of 2 was used for the next tests. The test fixture was heated to 330°F before pressure was applied. The pressure was increased above the dart in the test fixture to 15,000 psi and held for 3 hrs. 15 min. [Figure 10](#)

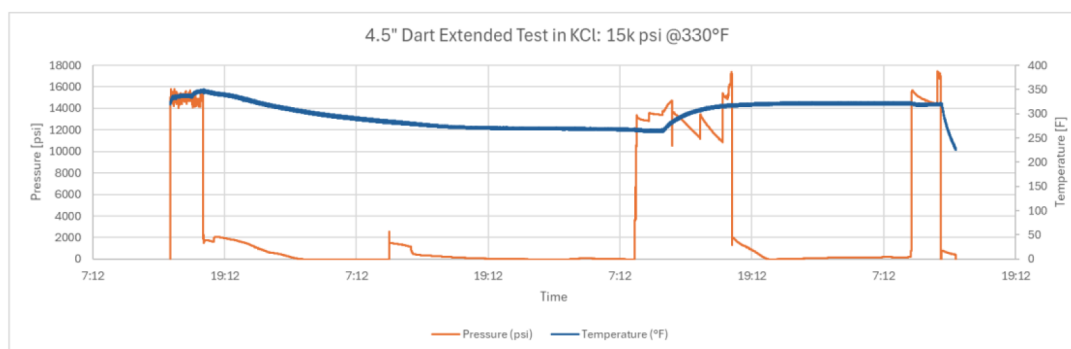


Figure 10—4.5" Dart Extended Test in Heavy Brine Material 2

Initially the IDD was to soak in the test apparatus with the heavier brine for 38 hours at 330°F, however the part of the heating element malfunctioned causing the temperature to drop from 330°F to 270°F. Pressure was increased to approximately 13,500 psi and held between 12,000 - 13,000 psi for 3 hours. After repairing the heating element, the pressure increased to 15,000 psi and held for 30 min, then was increased to 17,000 psi for an additional 15 min. A decision was made to let the IDD remain in the heavy brine for an additional 16 hours at the elevated temperature of 330°F. After 16 hrs. 43 min the pressure was increased to 15,000 psi for approximately 2 hours then taken up to 17,000 psi for 15 min. After approximately 69.5 hours with the IDD immersed in the heavy brine with no leaks, the test was arbitrarily terminated and deemed successful.

Shearable Dart Retaining Ring

During the design review it was identified that delays on surface, potentially lasting up to 12 hours before pumping operations would begin, could negatively impact the performance of the unactivated dart. To address this potential issue another feature was developed to retain the unactivated IDD at a cooler depth of the well to ensure its operational functionality. An internal diameter (ID) restriction near the surface of the well was identified as a suitable location to retain the unactivated IDD until the frac pumpers could start pumping. A dissolvable ring was engineered to attach to the upper section of the IDD which would shear off into multiple parts at a predetermined pressure signature as per Figure 11. This would verify the IDD was not in a higher temperature section of the well. Testing was performed to verify the pressure required to shear the ring away from the IDD. When the dissolvable ring was sheared away from the IDD it was designed to break into smaller segments to be corroded and to quickly disappear when the fluid is moving through the well.

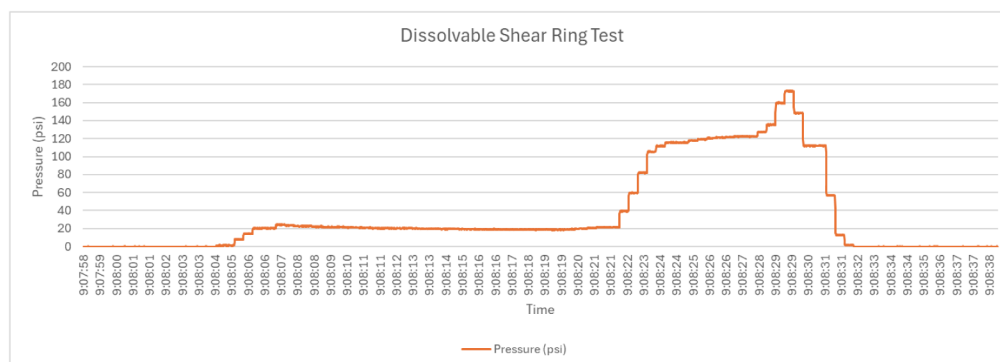


Figure 11—Dissolvable Ring Shear Test

Cement Wiper Dart

A cement wiper dart is a pumpable plug used in well operations to maintain separation between the cement slurry and other drilling or completion fluids during the cementing operation. Typically, the wiper

dart is launched from surface and pumped through the drill pipe and casing or liner, ultimately landing in a receiving landing collar. This action prevents the cement from mixing with both the internal and external fluids. Maintaining proper separation between displacement fluids and the cement slurry minimizes contamination, which is critical for ensuring strong cement bonding to the formation and achieving effective annular isolation.

The cement wiper dart generally consists of an inner non-flexible mandrel and with flexible fins made of rubber or a neoprene variant. These darts are available in various configurations and lengths, with fins designed to maintain continuous contact with the internal diameter (ID) of the casing or liner. This contact is essential to ensure the cement is displaced properly and for removing residual cement stringers or debris, ensuring a clean internal surface before the stimulation operation. Properly displaced cement along with clean casing ID ensures the ME MSF sleeves can be opened and prevents unwanted debris from being inadvertently pumped into the formation, which could otherwise plug formation pores. Upon landing in the collar, the dart generates a noticeable pressure increase, confirming that the cement slurry has been fully displaced and the operation is complete.

The ME MSF frac sleeves are designed with inner sleeves that shear down to open under pressure. Therefore, it is essential to test any cement wiper dart at surface with ME MSF sleeves to ensure it does not inadvertently activate or partially activate them during cement displacement operation. As part of the standard procedure, a detailed dimensional review of each wiper dart is conducted. Particular attention is paid to the hard outer diameter (OD) of the mandrel and the stiffness of the flexible wiper fins near the mandrel.

A key question in the evaluation is whether the wiper fins can fold back sufficiently and remain flexible enough to pass through the ME MSF sleeves without causing a significant pressure increase. Wiper darts with large OD mandrels and stiff fins may be disqualified before surface testing begins due to their inability to flex. When the wiper dart passes through restrictions in the casing or liner, some pressure increase is expected. These pressure spikes must be well understood and compared against the shear threshold of the inner sleeves to avoid premature activation.

Once a wiper dart passes the initial dimensional analysis, it is pumped through ME MSF sleeves at operational cement displacement rates. A data acquisition system (DAS) is used to record pressure and flow rate as the dart passes through the sleeves. The test sleeves are equipped with shear pins identical to those used in an actual wellbore deployment, to ensure test conditions accurately represent field conditions. During testing, darts are pumped through multiple ME MSF sleeves until pressure signatures stabilize. Typically, the first few passes result in higher pressures, but with repeated pump through runs, the pressure profile stabilizes at a lower, more consistent value. [Figure 13](#)



Figure 12—Cement Wiper Dart Test Assembly

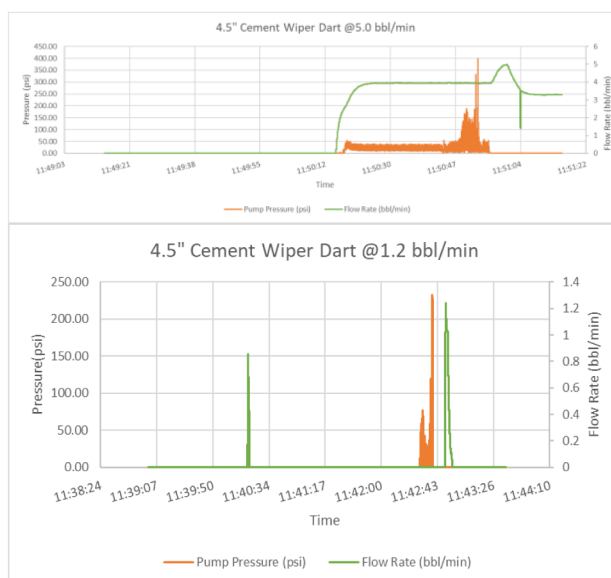


Figure 13—Initial Cement Wiper Dart Pump Thru with Various Rates

Each time the wiper dart passes through the end of the cement wiper dart fixture at various rates, the fins are inspected for any visible damage. If the wiper dart has no visible damage and the pump thru values are consistently lower than the shear value rating of the ME MSF sleeves, the cement wiper dart is qualified.

Deployment of the Technology

The candidate well for the innovative ME MSF system was selected to replace a cemented liner in a carbonate reservoir. Notably, the ME MSF system did not require any changes to the liner installation nor cementing procedures. The open hole size was 5%", approximately 4,100 ft in length and programmed for 11 stages with 10 of them utilizing the ME MSF system. Each stage, except for the 11th, had three ME sleeves each with six tungsten carbide nozzles. The landing sleeves had no nozzles and the 11th stage included two ME sleeves, for a combined flow area of 2.26 in² for the stages below the 11th stage. Although typical ME MSF cemented installations have nozzles in the landing sleeve, these nozzles were intentionally omitted for this installation due to the high acid concentration of the stimulation fluid and the potential exposure of IDD to dynamic conditions.

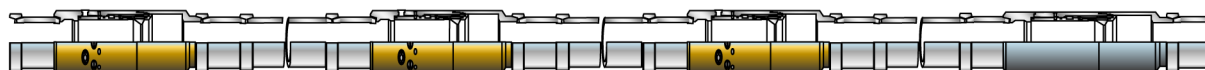


Figure 14—3 LE MSF Sleeves and 1 MSF Landing Sleeve

The last few liner joints required rotation and circulation to get the liner to the planned depth. Once the liner was on bottom the liner hanger was activated allowing the cement operation to commence. The cement wiper plug was pumped to the landing collar, allowing the wellbore to be pressured up to confirm pressure integrity of the liner. As the cement wiper dart testing had shown and field operations verified, the wiper plug did not interfere with the ME MSF sleeves as it was pumped through the liner when displacing the cement. Positive and negative pressure tests were successfully conducted as per [Figure 15](#)



Figure 15—Cement Pumping Chart

Stimulation Phase

Once the well completions were deployed and cemented successfully, the well was scheduled to planned stimulation operations. Acid fracturing treatment was chosen as stimulation technique for this tight carbonate/dolomite gas reservoir to increase effective conductivity between the formation and wellbore. The acid pumping schedule was designed to ensure an effective reservoir stimulation using standard dual-viscosity system & diverting agent at pumping rate between 25-50 bbl/min.

The functionality of ME MSF sleeves and IDD's for each stage during the fracturing job was clearly proven through the pressure signatures and surface ultra-sonic sensor. The opening pressure of all frac sleeves were expected and within design pressure range. The 1st stage of the fracturing operation was perforated after that, the frac sleeves in the next stages were opened via dropping corresponding IDD from the surface.

Summary of sequence of events during stimulation treatment through MSF as follows:

1. First stage:
 - Perforation using CT. In this case, it was conducted with the rigless site
 - Initiating the injection test and followed by main frac.
2. Second stage and the subsequent stages (frac sleeves):
 - Drop the IDD to open the ME MSF sleeves and isolate the previous compartment.
 - The injectivity test and main frac treatment are conducted through the opened sleeve.
 - Same IDD with adjusted programming is used to shift the corresponding sleeve(s) for next stage and performing the injectivity test and main treatment and repeat the same for the remaining stages.

To initiate the stimulation operations, first, a 2" coiled tubing with a jet nozzle assembly was run to displace the lower completion fluid to treated water. Once the displacement was completed, a correlation pass was done to position the perforation guns at the desired depth (20 ft) and interconnect the formation with wellbore in the Frac stage# 1. With a maximum of 8,500 psi surface pumping pressure with a rate of 2.5 bbl/min, an acid pill concentration of 20% HCl was injected into the first stage reducing the surface pumping pressure to 2,668 psi and rate to 5 bbl/min.

The frac pumps were lined up to the well and stimulation operations commenced with the first stage pumped through the perforation interval. After the completion of the first stage, the frac sleeve was opened by dropping the first IDD from the surface. Treatment fluid was injected as part of the second stage into the formation.

The IDD isolated the previous zone (first stage) and allowed opening of the three (3) frac sleeves to access the second zone for second stage. Subsequently, each zone was stimulated separately after isolating the previously stimulating zone by repeating the process of dropping identical IDD's programmed to correspond to their target sleeves.

As the IDD system requires power for the programmable activation, it will be affected by extreme well temperatures. To address this, a Shearable Dart Retaining Ring was added to set the IDD onto the upper flow coupling, located at a shallow depth as it was described in the previous sections, please reference to Figure 14 and 18. At this depth, the device will stand by for extended period of time only being exposed to low temperatures (83-85°F).

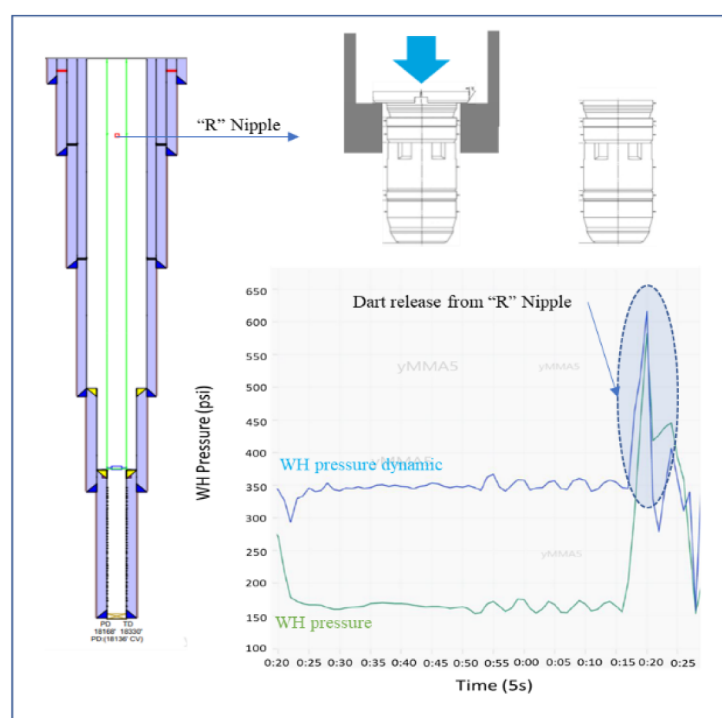


Figure 18—Shearable Dart Retaining Ring operational sequence

This requirement was incorporated because of the Wellhead Isolation Tool (WHIT) which is necessary for standard acid fracturing practices. These tools are designed to isolate the wellhead, ensuring the integrity and safety of the well during the high-pressure injection of fluids required in the treatments. The primary purpose of these tools is to isolate the wellhead and protect the well's components from the harsh conditions associated with frac treatments, such as extreme pressures and corrosive substances.

After the WHIT is rigged up and pressure-tested, a minor differential pressure of 450-650 psi will release the IDD from the upper flow coupling, to travel through the liner before activating to open the pre-programmed frac stage.

The fracturing operations proceeded smoothly throughout all stages. During the well stimulation, a zonal isolation analysis was conducted, confirming the IDD achieved good zonal isolation, thus preventing any communication between stages during the acid stimulation treatment. This analysis relies on the three following criteria to be met to verify the isolation between the compartments:

- Pressure difference during opening sleeves: A noticeable change in pumping pressure before and after seating the IDD in the landing sleeve at constant rate, confirmed the opening of the frac sleeves and the exposure of new formation.
- Pressure signature difference: Distinct pressure signatures between the instantaneous shut-in pressure decline after the injection test of one stage and the shut-in pressure decline of the earlier stage indicated no communication across formation between the new zone and the stimulated one.
- Absence of abrupt pressure change: The nonappearance of sudden pressure changes during the main stimulation treatment could be referred to maintained zonal isolation.

These key indicators collectively ensure that the zonal isolation and acid frac stimulations are effective, allowing for targeted and efficient stimulation of the desired zones. The three criteria used to confirm frac sleeves opening and zonal isolation are further illustrated through detailed charts and graphs, which display data collected during various fracturing stages of the first deployment with the technology. The data clearly supports the achievement of stage acid fracturing treatment, aligning with the aforementioned criteria. One fracture stage will be discussed to demonstrate how all criteria were achieved.

Open Crown Cap and dropped Dart 1, open Crown Valve (CV) and the dart landed on Hydraulic Master Valve (HMV), closed CV, equalized the pressure and opened HMV to launch the dart into the wellbore and to land on Flow Coupling. Install wellhead transducers, rig up Wellhead Isolation Tool (WHIT), connect frac lines and performed a pressure test. Start pumping slowly and saw indication of Dart ring shearing from Flow Coupling, continue chasing down Dart 1 till it successfully opened all of the clusters and landed in the landing sleeve at with a calculated volume of 241.7 bbl.

Performed an acid frac treatment and pumped the following volumes; 16,997 gal of Treated Water, 10,415 gal of Linear acid HCl 26%, 16,858 gal of 35lb/1000 gal crosslinked guar polymer, 15,354 gal of Emulsified HCl 26%, 2,548 gal of Linear Gel 35#, 980 gal of Tank Bottoms, and 1,200 lb of Diverter agent (462 gal) as per [Figure 22](#).

As evidenced in [Figure 19 and Table 1](#), the first condition for attaining zonal isolation was met, where a notable difference in pressure was observed before and after the Dart lands on the seat. The graph below illustrates the shift in differential pressure profile prior to and following the opening of the frac sleeves, while maintaining a consistent injection rate. The injection rate was constant at 12 bbl/min prior to shifting the multi entry frac sleeves during injectivity test of stage 2 to clearly show, the fluid was injected into the previously stimulated zone, stage 1. With a stable BHP of 11,000 psi there was a gradual increase of pressure reaching three BHP peaks ranging between 12,500-13,000 psi where the cluster sleeves were opened, followed by a drop in pressure to 12,200 psi, indicating injection into a new formation with no communication issues between the two zones. This observation indicates that the IDD has effectively isolated the zones between Stages and the stimulation treatment has accessed new rock in the new stage. Moreover, [Figure 20](#) will validate the hydraulic signals through an independent noise records of the multiple sleeve opening and landing of the IDD.

Table 1—Treatment summary of Stage 2

DART PARAMETERS AND OPERATIONAL SEQUENCE									
Dart Activation/Landing		Dart Shearing Through FC		Dart Chasing/Pumping					
Dart SN#	110853CTD-0000011	Shear Indication Pressure (FC)	Yes	Sequence#1	20 BPM	100 bbls			
Dart Name	Dart 1B	Shear Indication Acoustic (FC)	Yes	Sequence#2	15 BPM	60 bbls			
Dart Activation	3/17/2025 @ 21:23	Shear Pressure (FC) PSI	280	Sequence#3	12.2 BPM	86 bbls			
Dart Dropping	3/17/2025 @ 22:28	Fluid	TW	Total Volume	241.7 bbls				
Landing Indication at Flow Coupling (FC)	Yes	Flow rate (BPM)	3	Dart Landing	Successful				
MAIN FRAC PARAMETERS									
Pressure and Flow Rate Summary									
Variables	Max Press (PSI)	Max Rate (BPM)	Max BHP	Avg. Press (PSI)	Avg. Rate (BPM)	Surface ISIP (PSI)	Surface ISIP @ 5 min. (PSI)	Surface ISIP @10 min. (PSI)	Surface ISIP @ 15 min. (PSI)
Injection Test	7,947	20.2	13299	5,400	14	6,107	3,749	3064	2,603
Main Frac	13,872	37	17216	9,878	21.4	7,293	5,202	4202	3,575

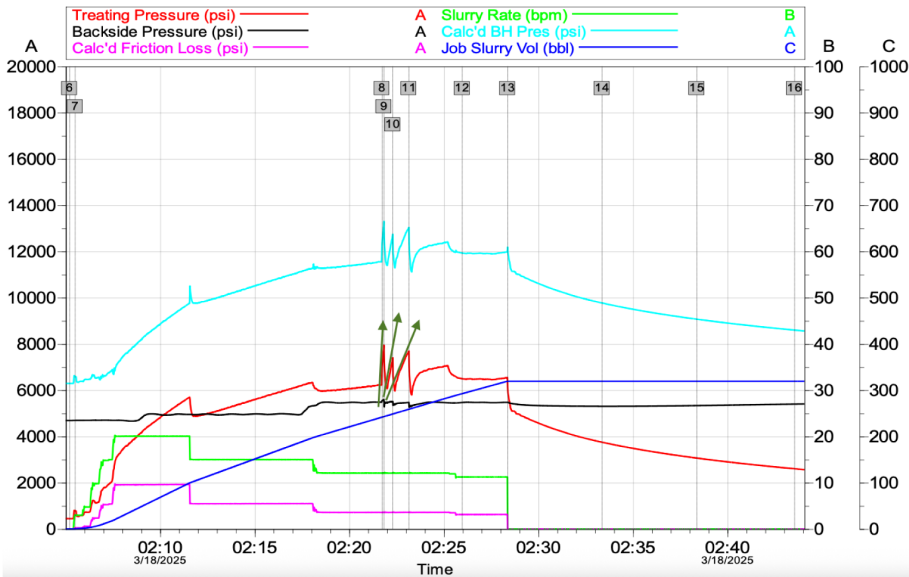


Figure 19—Different pressure response at same rate after opening the frac sleeves indicates positive isolation & opening of three hydraulic signals of MP frac sleeves

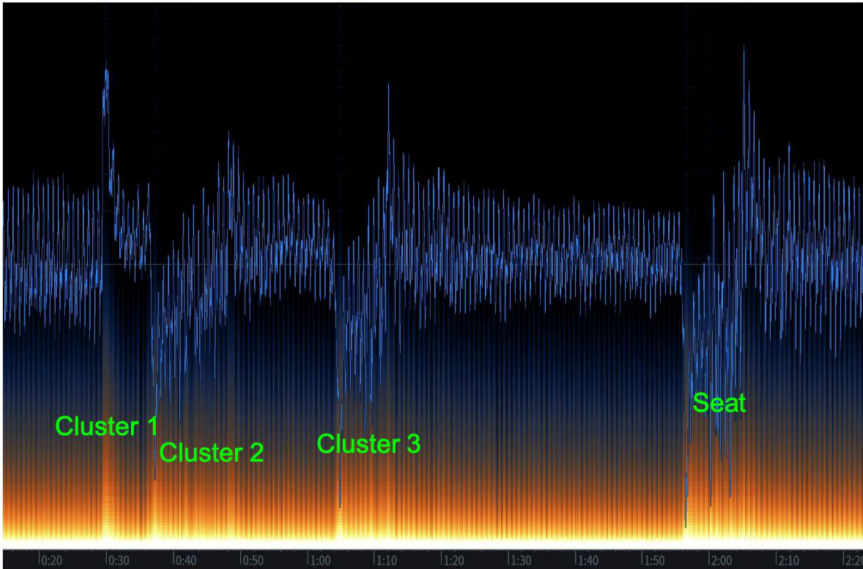


Figure 20—Three noise signals of MP frac sleeves

The second indicator of successful zonal stimulation is evident in the ISIP trend. Stage 2 was extensively analyzed pumping two injection tests, both injection tests were lower than and post-acid treatment of previous stage ISIP as shown in Figure 21 and Table 2. This disparity suggests that there is no communication between the two stages, thus confirming effective zonal isolation and stimulation between frac stages.

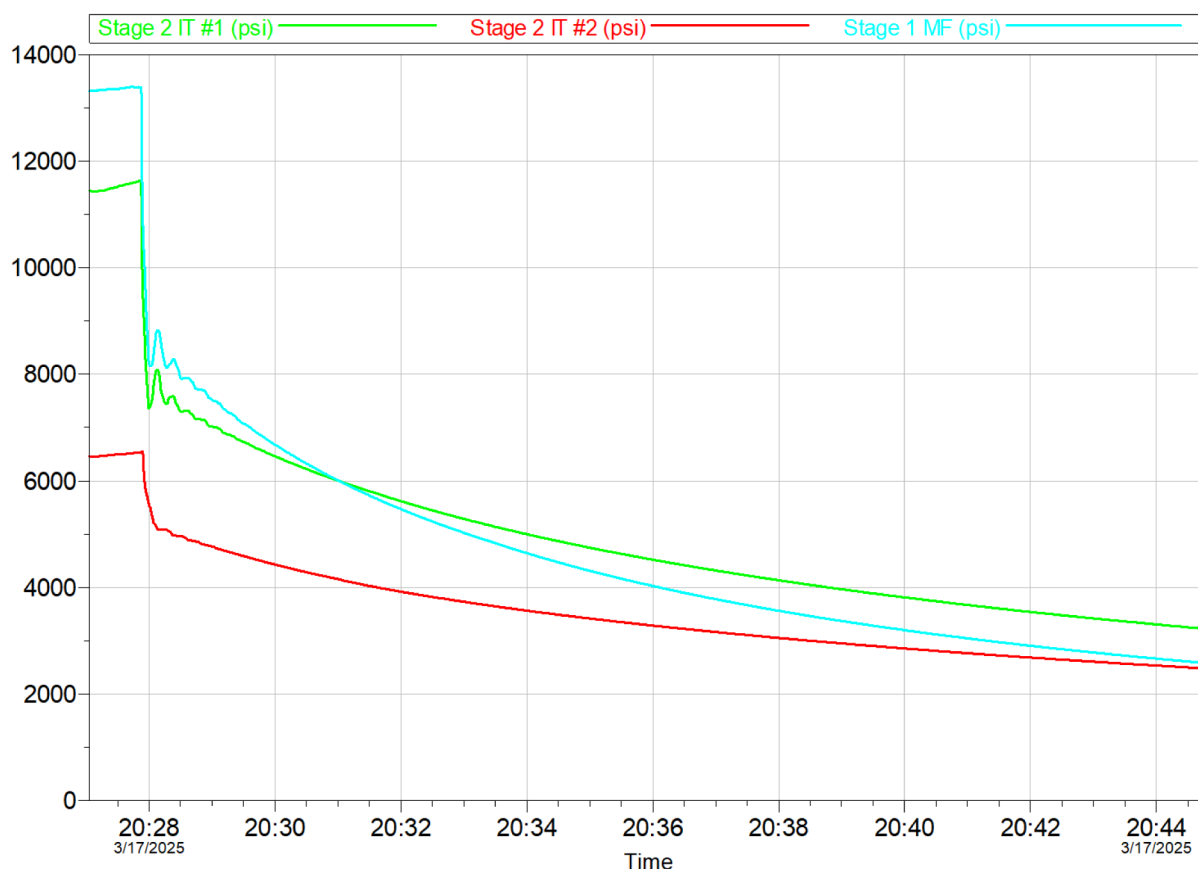


Figure 21—Different pressure signatures between the shut-in pressure decline after injection test of the stage and after the acid treatment of the previous stage

Table 2—Pressure difference between Stage# 2 injectivity ISIP and Stage# 1 Main frac

	Average Pressure before Dart Activation	Average Pressure After Dart Activation	ΔP , psi
ISIP, psi	6,106	7,050	994 @ 12 bbl/min

	Stage 1 - Main Frac	Stage 2 - Injectivity Test	ΔP , psi
ISIP, psi	8,506	5,120	3,386

The third indicator is the absence of any abrupt pressure changes during the main treatment, which indicates maintaining zonal isolation during main acid treatment. Figure 22 shows. All these indicators were observed in all 10 stages of the well, indicating successful stimulation operations.



During the rig-less operations, the innovative ME MSF completion system exhibited high reliability and repeatability during stimulation cycles. Eleven stages were successfully pumped without any intervention requirement, through the IDD drop technique to opening the frac sleeves and accessing the corresponding stages. The new system demonstrated positive zonal isolation, withstanding more than 12,500 psi differential pressure against the IDD, while a total of 16,200 bbl. of treatment fluid was injected into the formation (including high concentration acid). Post-job analysis of each frac stage indicated competent stimulation was achieved by the new MSF completion system, confirming effective treatment distribution between the stages as per the well stimulation program.

Table 3—Main treatment frac stage results

	Stage 1 Main Treatment	Stage 2 Main Treatment	Stage 3 Main Treatment	Stage 4 Main Treatment	Stage 5 Main Treatment	Stage 6 Main Treatment
Avg Rate – bpm.	15.9	21.4	21	28	28.8	27
Avg surface Pressure – psi.	9,483	9,878	8,176	9,166	7,683	8,905
Max Rate – bpm.	29.7	37	40	44.1	48	47
Max Pressure – psi.	13,391	13,872	13,790	13,733	13,424	13,574
Max BH Pressure – psi.	18,000	17,000	16,000	15,210	13,962	15,532
Fluid Loss Control Fluid – lb.	400	1200	1000	1300	1600	1000
Strong Emulsified Acid – bbl.	172	366	364	356	376	377
Linear Gel fluid – bbl.	86	61	61	58	83	60
Cross link fluid – bbl.	119	401	347	362	435	412
Treated Water – bbl.	398	405	678	419	379	457
Strong Acid – bbl.	121	248	239	238	334	260
Total Slurry Pumped – bbl.	896	1480	1689	1432	1606	1566

Table 4—Continuation: Main treatment frac stage results

	Stage 7 Main Treatment	Stage 8 Main Treatment	Stage 9 Main Treatment	Stage 10 Main Treatment	Stage 11 Main Treatment
Avg Rate – bpm.	31.2	33	30.1	31.8	26
Avg surface Pressure – psi.	13,682	8,535	6,985	7,312	5,573
Max Rate – bpm.	50.5	45.6	51.8	53.5	50.3
Max Pressure – psi.	13,036	13,351	13,803	13,633	13,499
Max BH Pressure – psi.	13,900	19,000	13,800	13,731	15,520
Fluid Loss Control Fluid – lbs	1200	1200	1000	1200	800
Strong Emulsified Acid – bbl.	358	356	371	371	170
Linear Gel fluid – bbl.	62	57	64	71	58
Cross link fluid – bbl.	359	356	369	390	224
Treated Water – bbl.	402	785	358	628	367
Strong Acid – bbl.	243	241	274	257	254
Total Slurry Pumped – bbl.	1424	1796	1434	1717	1073

Flowback chemical tracers

Chemical tracers played a pivotal role in the evaluation of hydraulic fracturing (fracking) operations in context with MSF lower completion deployed, it served as a critical tool for understanding fluid movement and stage effectiveness. These tracers are substances mixed during the fracturing process to track fluid dynamics within the reservoir, providing insights into the efficiency and uniformity of each acid fracturing stage.

The chemical tracers used in this job identified water, liquid gas condensate and a certain extent dry gas for each of the 11 acid fracturing stages. Following the fracturing process, the well entered the flowback

phase, where fracturing fluids are returned to the surface. The presence of chemical tracers in the flowback fluids was measured using specific techniques of chemical lab analysis with the collection of samples on surface during scheduled intervals.

The detection of tracers in all stages during the early flowback phase indicates that each stage is contributing to the flowback, signifying effective fracturing across all intervals. In the [Figure 23](#), it is showing the distribution of condensate and water tracers from stage 1 to stage 11, indicating the effective conductivity achieved in all Acidfrac stages.

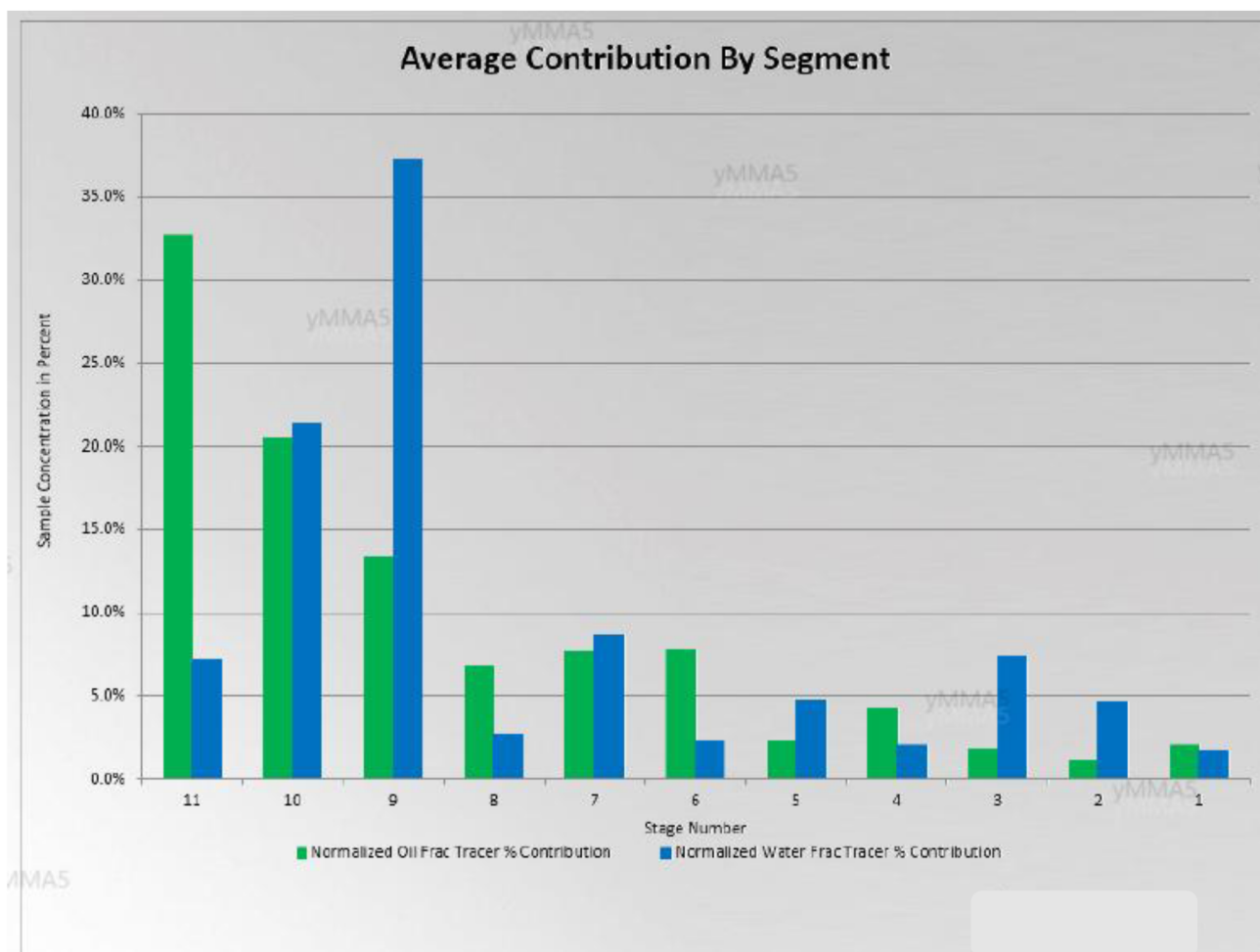


Figure 23—Water and Oil condensate chemical tracer distribution on post acid frac flowback.

Moreover, early flowback detection of tracers from Toe section only few hours after starting the operation indicated a partial dissolution of the IDD (flowing area) as per [Figure 24](#), enhancing overall operational efficiency. (IDD partial and total dissolution results were part of the qualification process, discussed briefly [Figure 12](#))



Figure 24—partial dissolution of IDD at flowback phase and expected final state of the dart after few weeks (Image available from qualification test)

Production performance analysis

The performance of Well-X, which features a limitless MSF completion technology, has been evaluated and compared to that of five offset wells (W-276, W-178, W-288, W-350 and W-297) within a 5 km radius. The offset wells, which target the same top porosity lobes of the carbonate reservoir, provide a suitable analog for assessing the efficacy of the new completion technology. The reservoir pressure in the region is characterized as normal, with no evidence of abnormal pressure conditions that could impact the comparison (the wells are tested between year 2024 and 2025). A normalized surface IPR (Inflow Performance Relationship) analysis of the well performance data, including production rates and wellhead pressures, has been conducted to quantify the benefits of the enhanced completion technology.

Figure-25 describes the completion types and number for acid fracture stages that were conducted on Well-X and its offsets. The completion types range from vertical well, open-hole MSF, plug and perf and cemented MSF with ball drop system. The frac stages ranges from 5 to 9 for horizontal wells, one stage for the vertical well, while Well-X is completed with 11 frac stages. In general, due to relatively good quality reservoir, the performance of the wells is satisfactory as it can produce commercially from any of these types of completion. As mentioned by [Rahim et al. \(2023\)](#), the improvement in gas production performance is expected with additional frac- stages. However, in Well-X, potential additional hydraulic fractures could be initiated and propagated from a geomechanical perspective, thanks to the limited entry technique combined with cemented conditions as frac-stage hydraulic isolation. As a direct observation, we observed, in [Figure-26](#), that Well-X surpassed the offsets for stabilized gas rate.

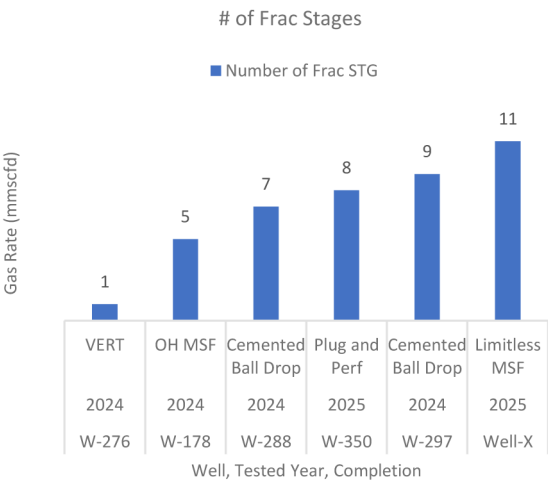


Figure 25—Gas production vs Number of Frac Stages

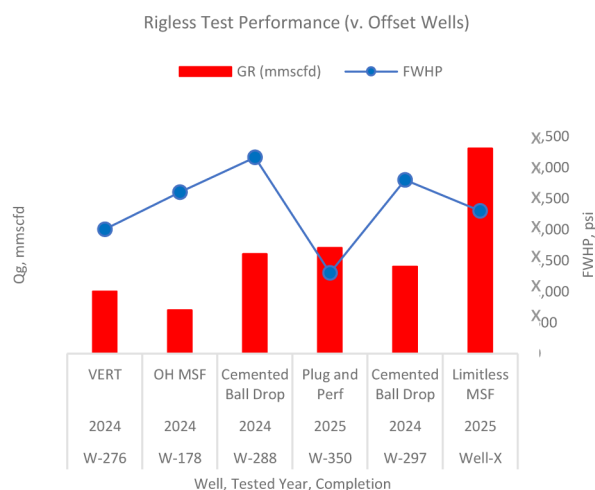


Figure 26—Rig-less Test Performance (v. Offset Wells)

A key aspect of this analysis is the normalization of gas production rates to a common wellhead flowing pressure, which enables a direct comparison of the wells' intrinsic productivity. The results of this analysis indicate that Well-X, equipped with the enhanced completion technology discussed in this paper, exhibits higher normalized gas production rate compared to the offset wells. This improvement in gas production can be attributed to the optimized feature of allowing more frac stages afforded by the new technology.

The performance of the rigless test for Well-X and its offsets are built into a surface IPR (qualitative comparison). As seen in Figure- 27, The wells AOF (Absolute Open Flow) vary broadly between all well in the sector. When the wells are normalized to produce at specific flowing wellhead pressure (FWHP) Figure 28, normalized performance of Well-X (completed with 11 stages) is 26% better compared to the nearest strongest well (W-297 that is completed with 9 stages). Comparing Well-X with a vertical well, W-276, Well- X yields nearly 4x fold of increase.

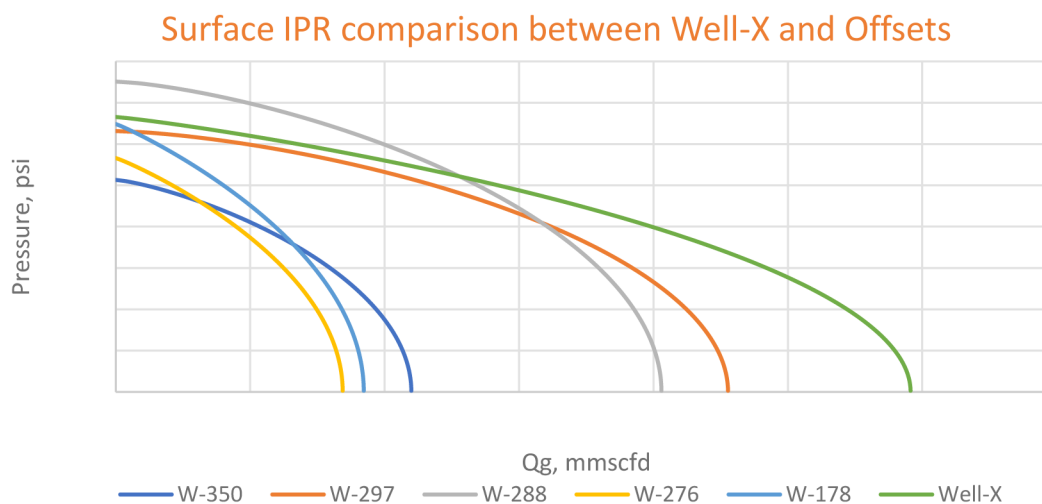


Figure 27—Surface IPR Comparison between Well-X and Offsets

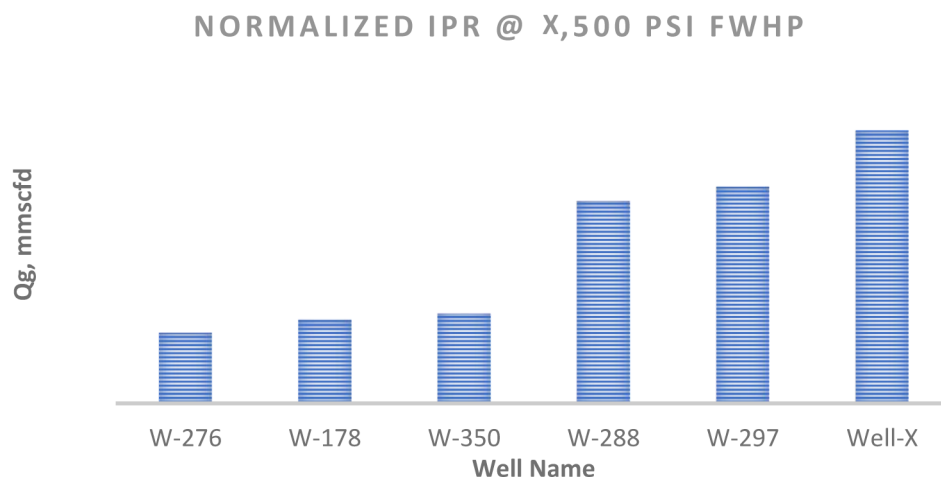


Figure 28—Normalized IPR Comparison between Well-X and Offsets

The comparative analysis of Well-X and the offset wells reveals distinct trends in their performance, with implications for completion strategy optimization. From reservoir quality point of view, there may be variations in performance that may be attributed to differences in reservoir properties, such as porosity and permeability, or the thickness (flow capacity) that can impact gas flow within the 5 km radius. Nevertheless, the enhanced completion technology in Well- X outperforms the offset wells, demonstrating its effectiveness in unlocking the full potential of the top porosity lobes in the carbonate reservoir with the implementation of a higher number of fracture stages. The findings of this study have significant implications for the optimization of completion strategies in the poorer reservoir quality where more reservoir contact will become an important factor to unlock the production of gas. A longer horizontal lateral with more numbers of frac stages can be implemented to achieve this objective.

Conclusions

This paper introduces a new high-pressure-rated (15,000 psi) cemented multi-entry MSF system and discusses its technical qualification, successful deployment, and results in a challenging HPHT extended-reach gas well. The paper also highlights the steps followed during the planning and execution of the completion deployment and acid fracturing treatment operations to successfully stimulate the ~18,500 ft HPHT gas well.

The improved design and operational capability of the new cemented ME MSF system greatly facilitated the completion deployment and overcame all the challenges observed in the extended-reach lateral. The new 15K cemented ME MSF system demonstrated high performance in both the deployment and stimulation phases, successfully adapting to a 15,000 psi pressure differential application and slim geometries without interfering with the cementing design and operations. In terms of flexibility, the new cemented MSF system does not require any custom wiper dart, as the ID of the frac sleeve is constant. Multiple tests were completed with standard approved wiper darts, proving compatibility with multiple liner hanger providers.

Detailed validation testing was performed on the new Intelligent Dissolvable Dart (IDD) before deployment. The results were substantial, demonstrating high integrity and acid resistance properties under high-pressure, high-temperature, and low-pH conditions. Additionally, the IDD showed full dissolution over time. Furthermore, during the stimulation phase, the completion system demonstrated effective zonal stimulation and isolation, withstanding high differential pressures of more than 12,500 psi across the MSF stages. This proved that the IDD maintained high-pressure integrity even with prolonged contact with low-pH, high-salinity fluids, and high temperatures.

Rig-less stimulation operations were successfully completed by stimulating eleven (11) acid frac stages according to the stimulation program. A detailed post-job analysis for all stages was performed after the

completion of stimulation operations to ensure stimulation efficiency and reliability using chemical tracers (water, oil, and gas) and well production evaluation in comparison with offset.

References

- Zillur Rahim; Mark Watson; Stuart Wilson. Pushing the Limits of Multi Port Open Hole Completions to Maximize Hydraulic Fractures. Paper presented at the SPE Annual Technical Conference and Exhibition, October 2023. Paper Number: SPE-215172-MS. <https://doi.org/10.2118/215172-MS>
- Rasim Serdar Rodoplu; Adegbenga O. Sobowale; Jon E. Hanson; Beau R. Wright. 15K Multistage Fracturing Completion System Addresses Operational Challenges, Improves Production Performance and Cost Savings in Open hole Environments. Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November 2021. Paper Number: SPE-207368-MS. doi: <https://doi.org/10.2118/207368-MS>
- Rommel Arias, Ricardo Bernal, Muzzammil Shakeel, Marc Normand. Successful Deployment of Acid-Resistant Dissolvable Frac Ball for Multi-Stage Acid Stimulation Application in an Offshore Multi-Lateral Well. Paper presented at the ADIPEC, October 31-November 3, 2022 doi: <https://doi.org/10.2118/211761-MS>
- Gongbo Long; Guanshui Xu. The Effects of Perforation Erosion on Practical Hydraulic-Fracturing Applications. *SPE Journal* **22** (02): 645–659. <https://doi.org/10.2118/185173-PA>
- Jacobs, T. 2021. Oviniv: Reliable Data, Better Plug Choices Solve Casing Erosion and Improve Stimulations in Shale Wells. *Journal of Petroleum Technology, JPT*. <https://jpt.spe.org/ovintiv-reliable-data-better-plug-choices-solve-casing-erosion-and-improve-stimulations-in-shale-wells>.
- E. R. Davis, J. J. Constantine, M. L. White, M. M. Ferris, M. J. Watson, M. M. Schinnour, N. R. Zwarich, M. E. Woodard, K. N. Metzgar, M. Nozaki. "A Shift or Not a Shift?" that was the Question: Robust Qualification and Testing of Available "Unlimited" *Frac Sleeve Systems Yield Varied Results*. Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, June 13-15, 2023. Paper Number: URTeC: 3849448. <https://doi.org/10.15530/urtec-2023-3849448>